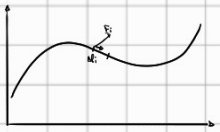
$|F_i| \cdot |\Delta l_i| \cdot \cos\theta$

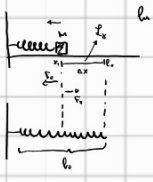
$L_\gamma(\underline{F}) = \sum_i F_i \cdot \Delta l_i = \int_\gamma \underline{F}\, d\underline{l}$ $\qquad$ $\int_\gamma (\underline{F}^{(ap)}) = T_B - T_A$

$T = \frac{1}{2} m v^2 = \frac{1}{2} m \underline{v} \cdot \underline{v}$

Energia cinetica



Lavoro fatto su una molla



La lunghezza a riposo della molla sia l_0

La compressione vale $\Delta x = |\Delta x|$.

Sia k la costante elastica della molla.

$\underline{F}_0$

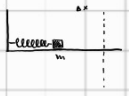
$F_e = -kx \rightarrow x \le l_0 \rightarrow F_0 = -F_e$

$F_e = k(l_0 - x)$

$L_\gamma = k\int_0^{x_1} (l_0 - x)\, dx = k\left[\frac{(l_0-x)^2}{2}\right]_{x_0}^{x_1} =$

$= -\frac{(l_0-x_1)^2}{2} \cdot k = \frac{-k\Delta x^2}{2}$

↓ il lavoro compiuto nella compressione di una molla di Δx con costante elastica k



$L = \int_{x_0}^{x_1} F(x)\, dx = \int_{x_0}^{x_1} \bar{F}_1 \left(1 + \frac{1}{6}\frac{x^2}{x_1^2}\right) dx = \bar{F}_1 \left[x + \frac{1}{6}\frac{x^3}{x_1^2}\right]_{x_0}^{x_1}$

$L = \frac{1}{2} k \cdot \Delta x^2 = T_p = \frac{1}{2} m v_0^2$

$v_0 = \sqrt{\frac{k}{m}}\, \Delta x$

$x(t) = \Delta x_0 \cdot \cos\sqrt{\frac{k}{m}}\, t$

↓ se ricavo ← derivando $x(t)$

$v(t) = \sqrt{\frac{k}{m}}\, \Delta x_0 \cdot \sin\left(\sqrt{\frac{k}{m}} \cdot t\right)$

Moto Circolare

. Strada Piana

1000 Kg $\quad R = 30\,m$

$v = 15\,\frac{km}{h} =$

$\mu_d = 0.6$

$\mu_s = 0.25$

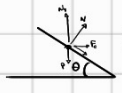


$N = mg$

$F_c = \frac{mv^2}{R}$

↑ forza centripeta

$\mu N = \mu mg \ge \frac{mv^2}{R}$



con a è nullo lungo y: $F_y = ma_y = 0 = -mg + N\cos\theta$

$N = \frac{mg}{\cos\theta}$

v, R, m

θ per fare in modo che non venga l'attrito

$F_c = \frac{mv^2}{R} = N\sin\theta = \frac{mg \cdot \sin\theta}{\cos\theta} =$

$= mg\tan\theta$

$\tan\theta = \frac{v^2}{Rg}$